

BCAC0030: MOBILE APP DEVELOPMENT-I

Objective: To introduce students to cross-platform mobile application development using Flutter and Dart, enabling them to build modern, responsive, and feature-rich applications for Android and iOS platforms.

Credits:03

L-T-P-J: 3-0-0-0

Module No.	Content	Teaching Hours
I	Introduction to Mobile Application Development – Overview of mobile app evolution, significance, and use cases across different domains such as education, health, and entertainment. Basics of Core Java – Understanding object-oriented programming principles, control structures, classes, and inheritance essential for Android development. Android Operating System Overview – Explore the architecture, key features, and version history of Android as a mobile platform. Android Development Environment Setup – Install and configure Android Studio, create and manage Android Virtual Devices (AVDs) and deploy apps to physical devices. Activity and Lifecycle – Learn the structure of Android activities, their lifecycle methods, state changes, and best practices for lifecycle management. Android Project Structure and AndroidManifest.xml – Analyze the components and structure of a typical Android project, including understanding the role of the AndroidManifest.xml file. UI Design Principles and Layouts – Apply UI/UX design concepts using different layouts such as LinearLayout, ConstraintLayout, RelativeLayout, and FrameLayout to build flexible user interfaces. Widgets and UI Components – Use standard UI elements like TextView, EditText, Button, and ImageView to build forms and interactions. Advanced Input Controls – Incorporate components such as RadioGroup, RadioButton, CheckBox, and Spinner to handle user selections and options. Dialogs and Lists – Implement interactive elements like AlertDialog, AutoCompleteTextView, ListView, GridView, and provide feedback using Toast and Snackbar messages.	19
II	Fragments – Learn how to create and manage fragments, understand their lifecycle, perform fragment transactions (add, remove, replace), and implement communication between fragments and their host activities for modular UI design. SQLite Database – Understand how to perform CRUD operations (Create, Read, Update, Delete) using SQLite, implement a database helper class, and follow best practices for local data storage and management in Android apps. RESTful API Integration – Explore the basics of web services, differentiate between REST and SOAP, use Postman to test API endpoints, and integrate web APIs in Android apps using libraries like Retrofit or Volley with proper handling of JSON responses. Firebase Integration – Set up and use Firebase services, including working with Firebase Realtime Database or Firestore to store and retrieve structured data, and implement secure user authentication using Firebase Authentication (email/password method). Background and Foreground Task Services – Learn to execute tasks in the background or while the app is in the foreground using Services, JobScheduler, WorkManager, and foreground services for long-running or periodic operations such as syncing data or downloading files.	19

References:

- "Flutter for Beginners" by Alessandro Biessek
- "Beginning Flutter: A Hands-On Guide to App Development" by Marco L. Napoli

Outcome: After completion of course, student will be able to:

CO1: Understand various mobile application architectures, design principles, and development patterns specific to Flutter.

CO2: Describe the components, structure, and functionalities of the Flutter framework and the Dart programming language.

CO3: Develop cross-platform mobile applications using Flutter and its core features.

CO4: Deploy, test, and debug Flutter applications on both Android and iOS platforms

Mapping of Course Outcomes (COs) with Program Outcomes (POs) and Program Specific Outcomes (PSOs):

COs	POs/PSOs
CO1	PO1,PO5/PSO1
CO2	PO1,PO3/PSO1, PSO2
CO3	PO1,PO3/PSO2, PSO4
CO4	PO1/PSO1